AQA GCSE Physics
Topic – Space Physics
Section 1: Detailed Revision Notes

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1. The Solar System

The Solar System consists of:

- The Sun (a star)
- Eight planets
- Dwarf planets
- Moons
- Asteroids and comets

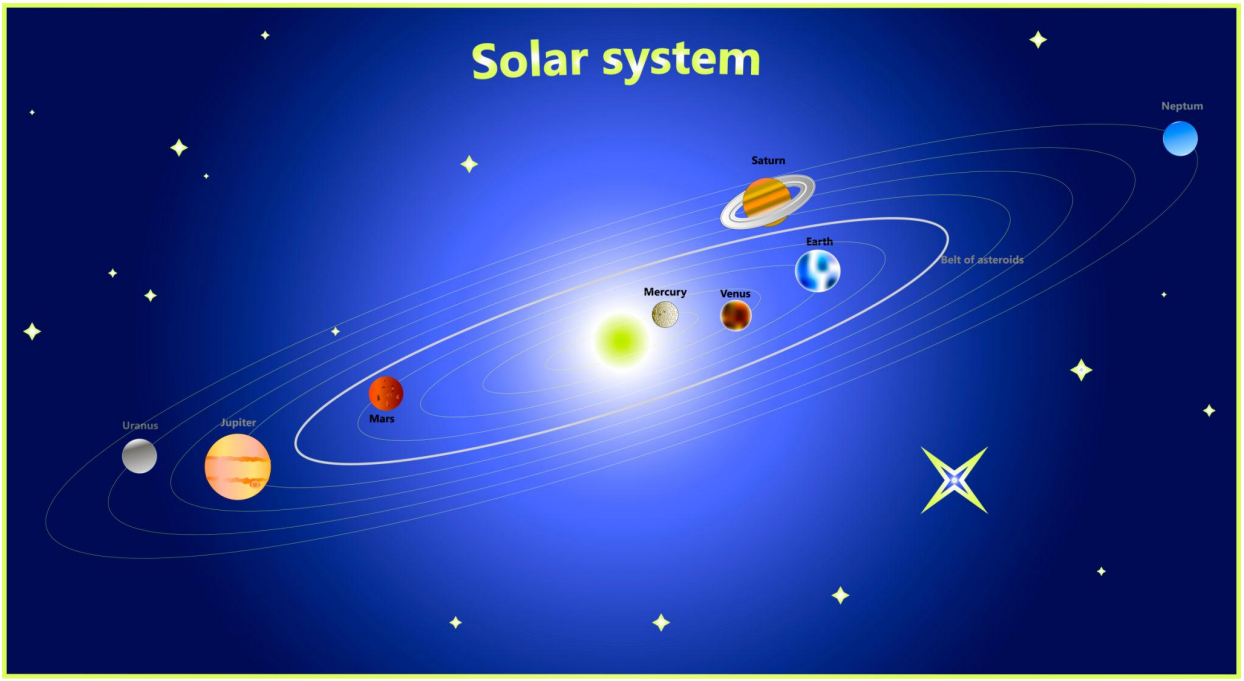
The Sun contains **99.8% of the mass** of the Solar System.

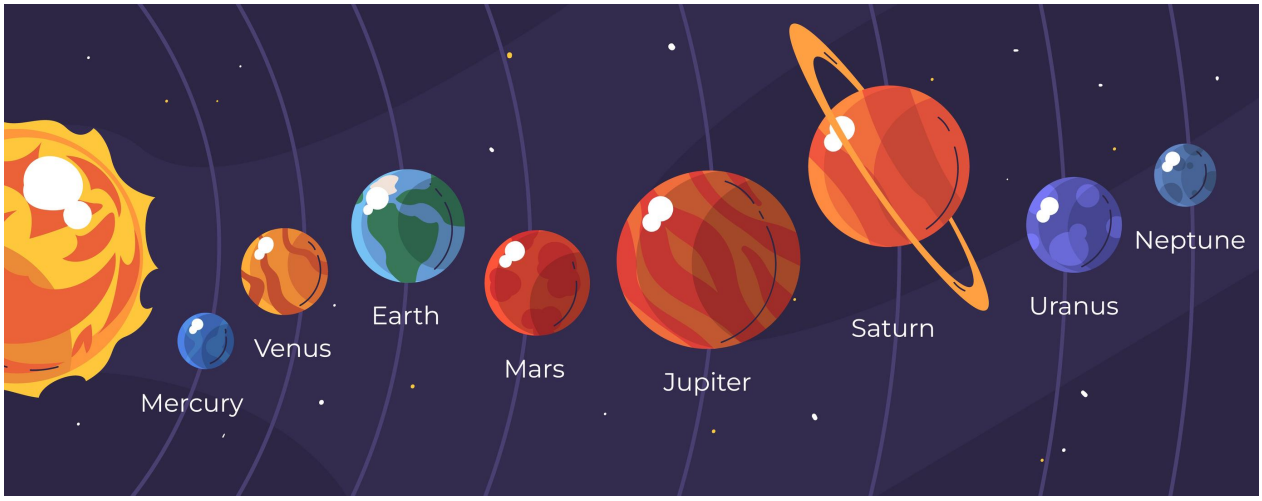
Order of Planets (Closest → Furthest from Sun)

Mercury
Venus
Earth
Mars
Jupiter
Saturn
Uranus
Neptune

Mnemonic:
My Very Easy Method Just Speeds Up Naming

Solar System Diagram





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2. Formation of the Solar System

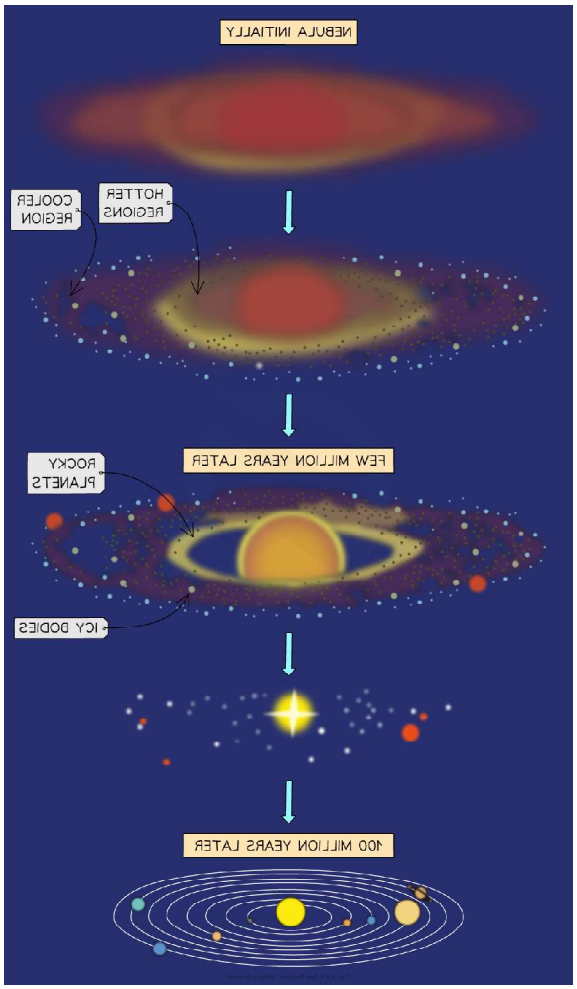
The Solar System formed about **4.6 billion years ago** from a cloud of dust and gas called a **nebula**.

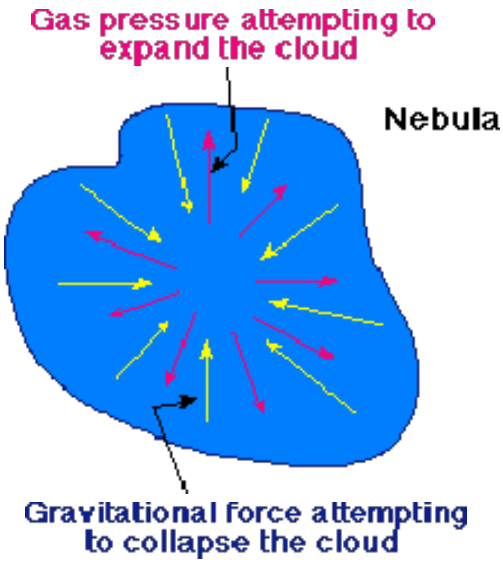
Stages:

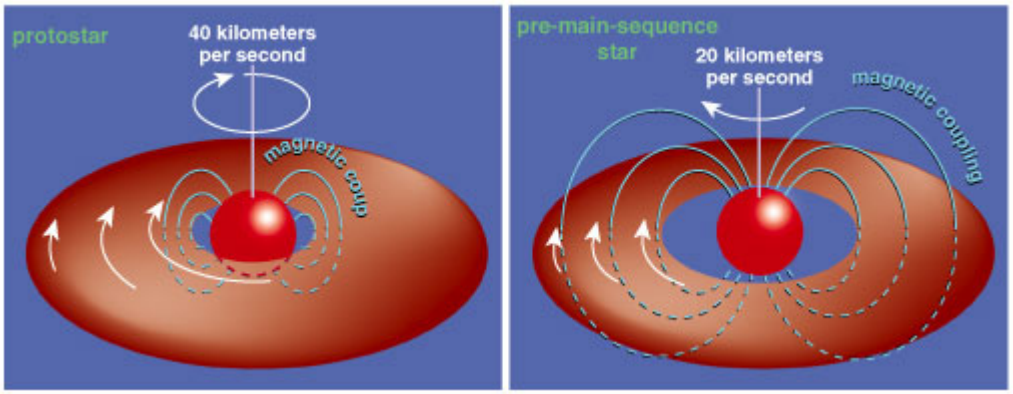
1. Nebula collapses under gravity

2. Forms a protostar (young Sun)
3. Planets form from remaining material

Nebula Formation Diagram







3. Life Cycle of a Star

Stars form from nebulae.

Main Stages:

Nebula → Protostar → Main Sequence Star

After this depends on mass.

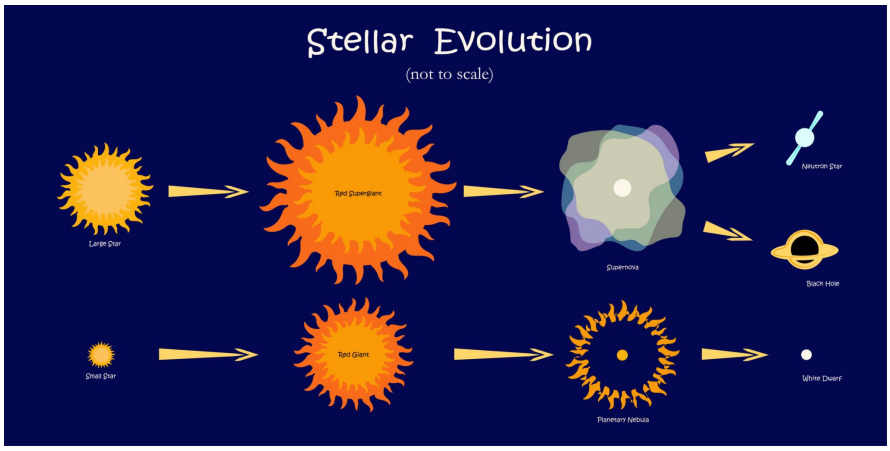
Low-Mass Star (e.g. Sun)

Red Giant → White Dwarf → Black Dwarf

High-Mass Star

Red Supergiant → Supernova →
Neutron Star OR Black Hole

Life Cycle of Stars Diagram



4. Orbits & Gravity

Gravity is the force that:

- Keeps planets in orbit
- Keeps moons orbiting planets
- Keeps artificial satellites in orbit

Orbital Motion

An orbit occurs because:

- Gravity pulls object inward
- Object moves forward due to inertia

This creates circular or elliptical motion.

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5. Satellites

Natural Satellites

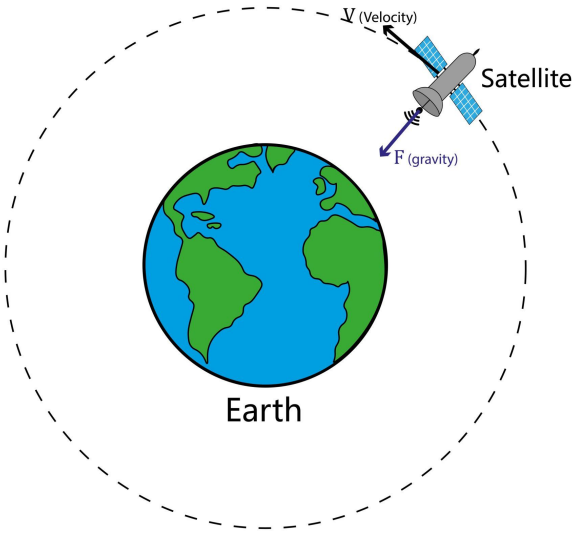
The Moon is a natural satellite of Earth.

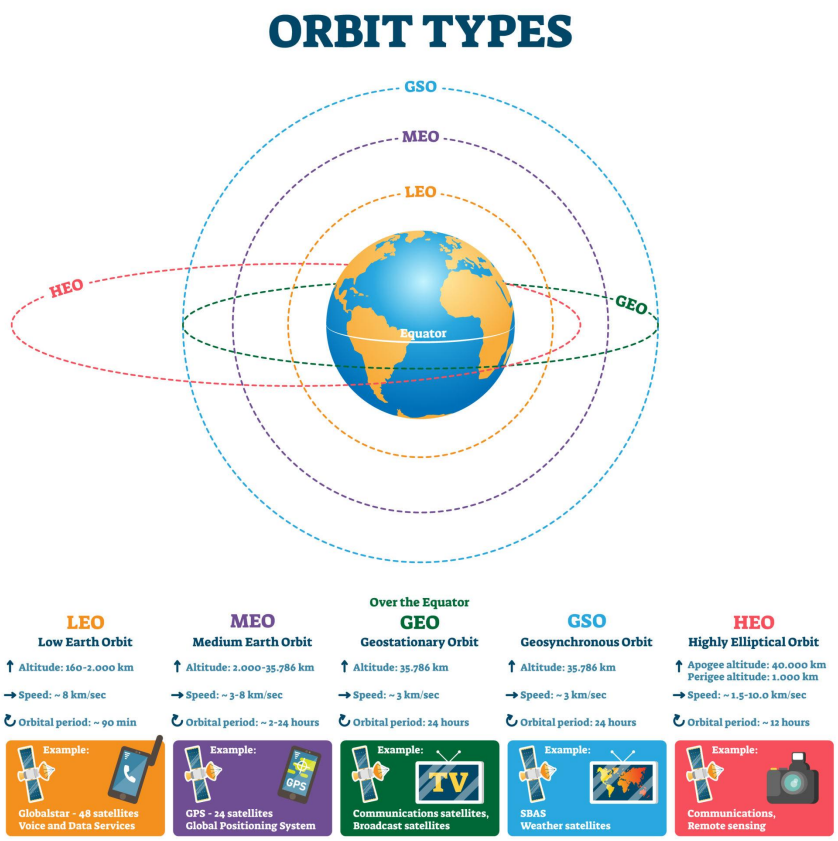
Artificial Satellites

Used for:

- Communication
- GPS
- Weather monitoring

Satellite Orbit Diagram





Geostationary Orbit

- Orbit above equator
- Takes 24 hours
- Stays above same point on Earth

6. Red-Shift & Expanding Universe

Light from distant galaxies is red-shifted.

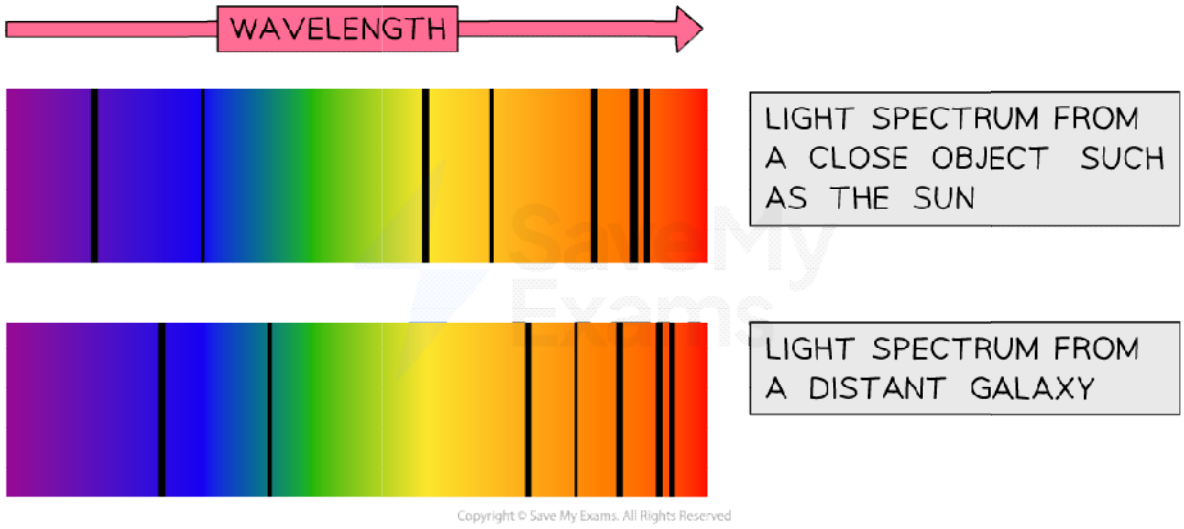
Red-shift means:

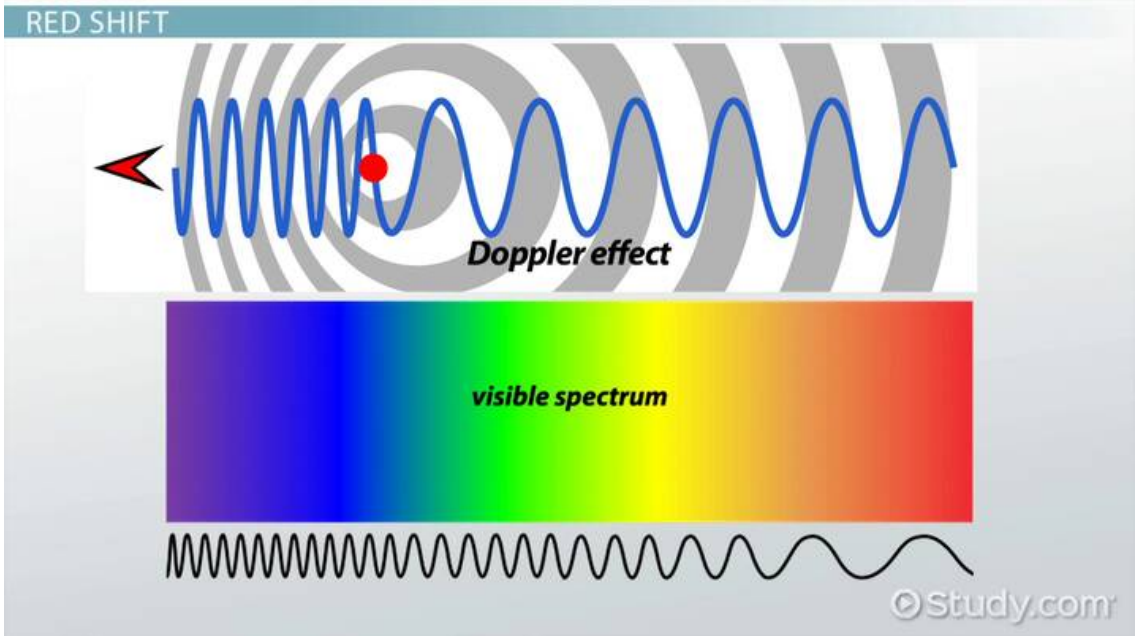
- Wavelength increases
- Light shifts towards red end of spectrum

This shows:

The Universe is expanding.

Red-Shift Diagram




RED SHIFT
Doppler effect
visible spectrum
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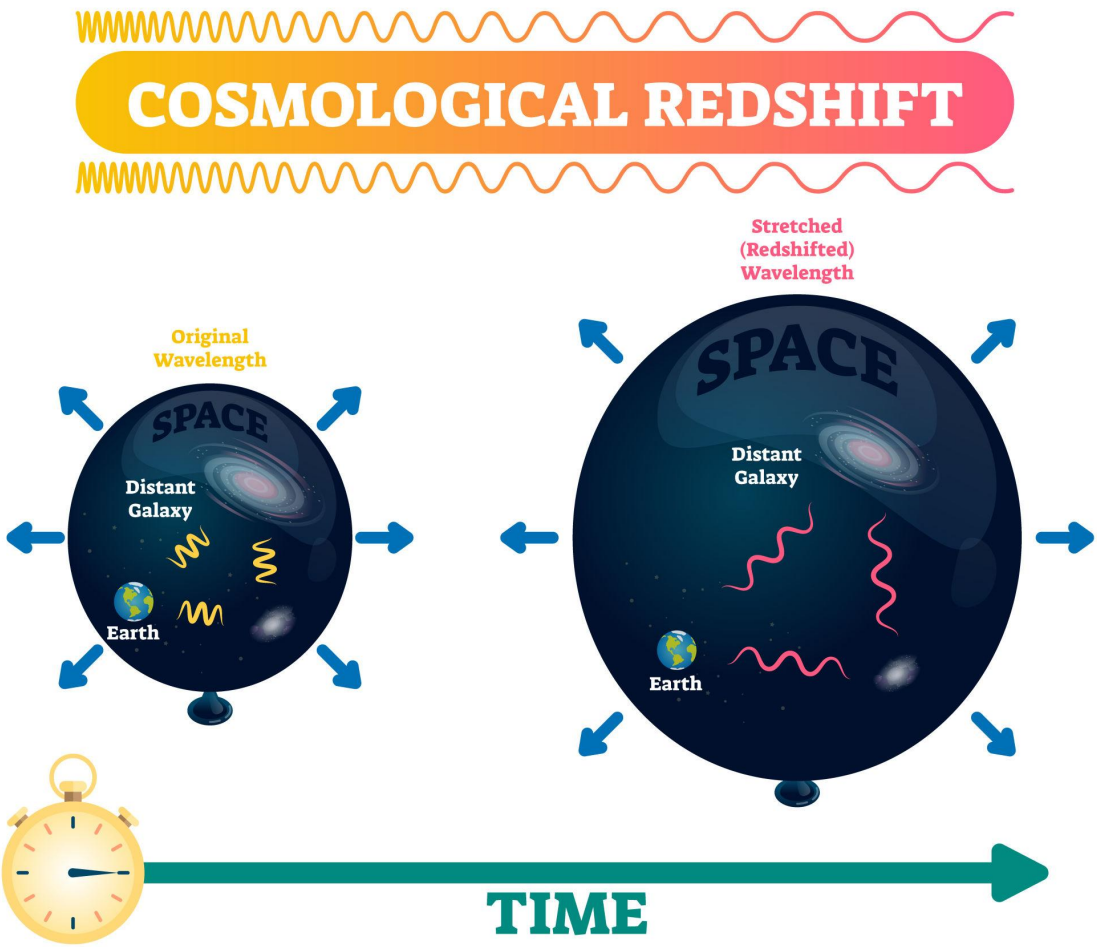
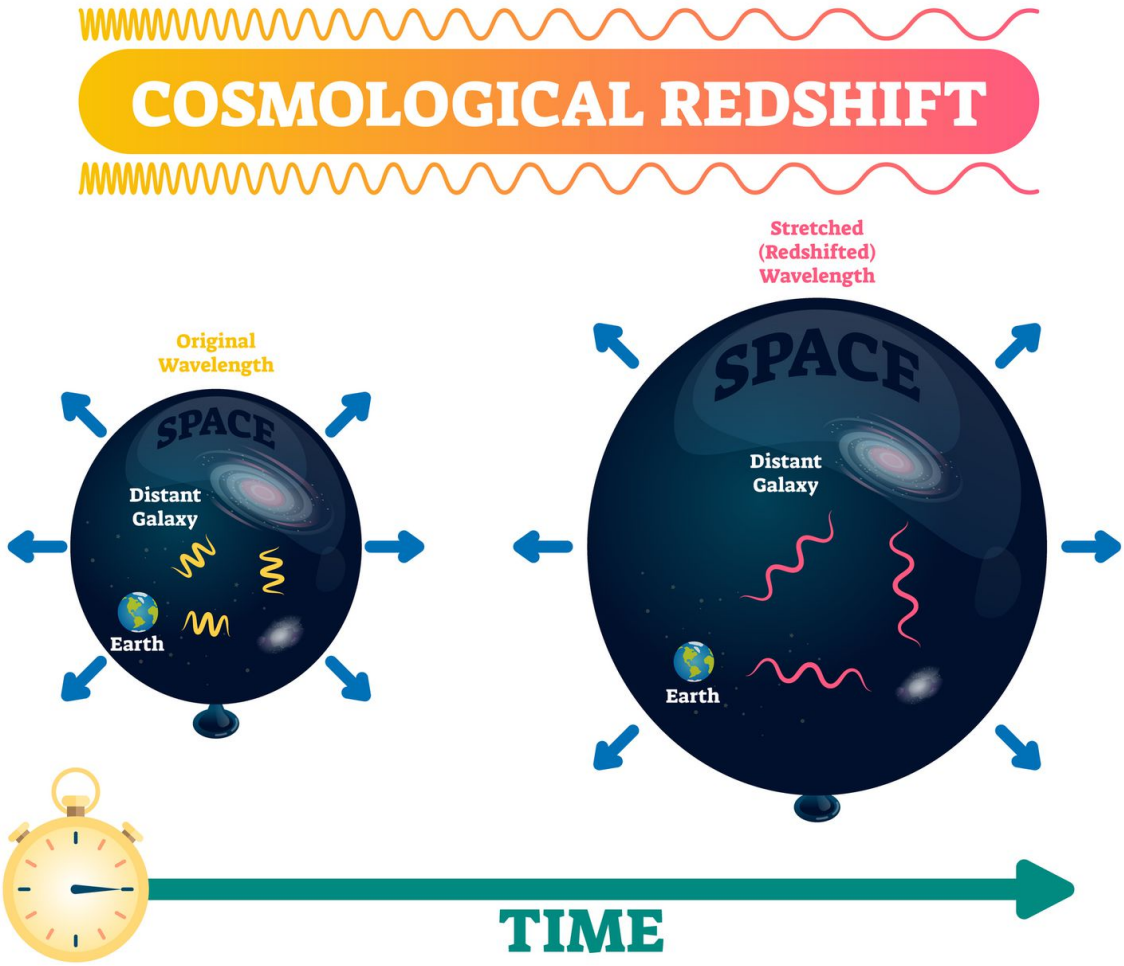

COSMOLOGICAL REDSHIFT
Original Wavelength
SPACE
Distant Galaxy
Earth
Stretched (Redshifted) Wavelength
SPACE
Distant Galaxy
Earth
TIME



Big Bang Theory

Evidence:

- Red-shift
- Cosmic microwave background radiation

The Universe began from a very hot, dense state about **13.8 billion years ago**.

7. Common GCSE Exam Mistakes

- Confusing rotation and orbit
- Forgetting order of planets
- Mixing up red giant and red supergiant
- Forgetting red-shift = moving away
- Writing gravity causes speed (it causes acceleration)

Key Definitions Summary

Nebula – Cloud of gas and dust
Main sequence – Stable star fusing hydrogen
Red shift – Increase in wavelength due to moving away
Orbit – Curved path due to gravity
Geostationary – Matches Earth's rotation

End of Section 1 – Space Physics